Making Profit in the Stock Market Using HMMs

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ABSTRACT
In this paper, I describe the process I used to achieve profit in the stock market over one year, trading ten stocks. I utilized Hidden Markov Models to predict each stock price daily. The Baum-Welch algorithm was used to train Hidden Markov Models for each stock using training data from 2009-2011. These Hidden Markov Models were then applied to stock data from 2011-2012 to attempt to make a profit. A profit of 26% was yielded from the testing of the Hidden Markov Model trading simulation.

Author Keywords
Hidden Markov Models, stocks, profit, python.

INTRODUCTION
The project faces the problem of trying to make profit in the stock market. This is a problem that concerns many people in the business world and anyone interested in making some potentially and hopefully easy money. To play the stock market with the hope of making a profit it helps to have an edge over the competition. One way to do this is to have a good way to predict future prices and trends. Future prices and trends can be predicted using older data that is then applied to the incoming data. For this project I was interested in exploring the ability of Hidden Markov Models to handle the prediction of future stock prices.

PROJECT DESCRIPTION
I built a system that can create Hidden Markov Models[1] based off of stock data input, use the generated Hidden Markov Models to predict prices, and run a simulation of trading stocks on a day to day basis. I created this system using Python, due to exposure of it during the class and its ability to quickly prototype an idea. Python allowed me to utilize high level programming capabilities such as easily opening and looping through a given set of files. Python also allows small changes to be made to the code and then the code can run without having to take the time of recompiling. Python's built in dictionary system also made it easy to keep track of stocks and their computed values.

The first part of my program involved analyzing ten different set of stocks from the years 2009-2011 and giving the values associated with the closing price of the stocks to my implementation of the Baum-Welch algorithm. The Baum-Welch algorithm was chosen for training due to the recommendation from [2]. This algorithm could take the stock values of the stock's price each day and compute a Hidden Markov Model that accurately represented the price trends seen during this time. My Hidden Markov Models had four hidden states in them which were "High increase", "Low increase", "Low Decrease", and "High Decrease". There were four possible observations to emit which were 1.0-0.5, 0.5-0.0, 0.0—0.5, and -0.5- -1.0, which were calculated by subtracting the closing price by the opening price then dividing by the opening price. Ten Hidden Markov Models were created, one for each stock.

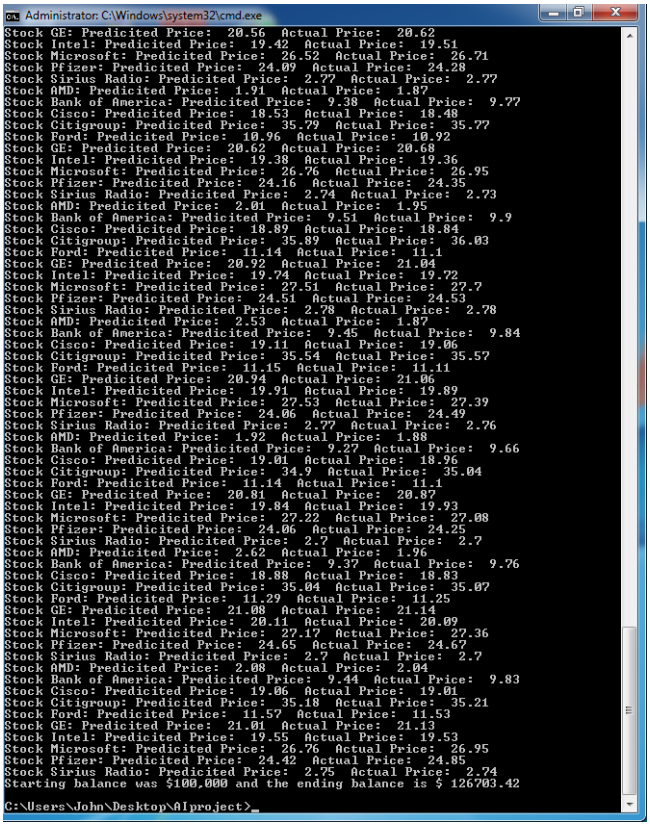

Figure 1 Example output of program.

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With a trained Hidden Markov Model, I now passed the values of the real data from 2011-2012 to be predicted on a daily basis and then buy stock based on whether the price of the next day would increase or not. The values from the 2011-2012 data were passed to the Hidden Markov Model to compute a likelihood value. This value is then compared to others generated from the test data. The closest likelihood value that is found in the test data is used as the point of prediction. The stock price change that occurred at that likelihood value is assumed to occur at this present price to attempt to predict the future price. If the price would increase, the program would buy the stock then automatically sell it the next day at whatever the close price ended up being, and if the price was going down the program would not buy the stock. This process was simulated over the course of a year using 2011-2012 data for ten stocks. Example 1 displays an example output of the program with the given data.

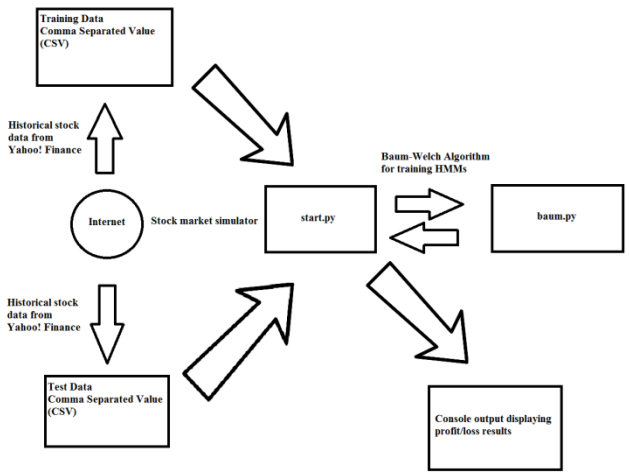

Figure 2 Technology diagram.

The stock data used is historical stock data retrieved from Yahoo! Finance[3]. The training data consisted of the years 2009-2011 and the real data is for the year period of 2011-2012. The ten stocks I attempted to trade for profit are AMD, Bank of America, Cisco, Citigroup, Ford, General Electric, Intel, Microsoft, Pfizer, and Sirius Radio. The columns contained in the csv files for the data are date, stock opening price, stock high price, stock low price, stock closing price, volume for a given day, and adjusted price. The main columns used in my program were the date, opening, and closing prices. Figure 2 displays how all the technologies came together in the project, including the datasets and their relations.

ANALYSIS OF RESULTS

The results of the daily stock trading were a 26%+ profit. I originally started with a bank account of $100,000 that then evolved into $126,703.42. These figures indicate that a profit is possible, and that a Hidden Markov Model is capable of predicting some price trends. The simulation model was simplified so the results may be a bit blurred. I also set in place a strict limit to buy no more than $3,000 worth of a stock that looked like it had the opportunity to be profitable, and always sold the next day if I bought a stock. Due to the fact that the system sold the stock every time after it predicted an increase and I did see a profit means that the system predicted an increase in price more often than not. My main metric for determining if there was success with the project was whether or not the project yielded any profit. In this case, a sizeable profit was yielded. I am pleased with my results, and believe that even though there were simplifications made to the trading algorithm, there is strong evidence that a Hidden Markov Model can predict stock market prices to a certain extent.

DISCUSSION

I learned a great deal about Hidden Markov Models from this project. I learned not only the theories and concepts behind Hidden Markov Models, but also how to implement one and a system to support it. I deepened my understanding of hidden states associated with Hidden Markov Models and the observations being tied to the states. Understanding this correlation aided me much better in understanding what an emissions probability matrix represented. I also learned about how to use a trained Hidden Markov Model in determining values for data and what those values signify.

I learned about the Baum-Welch algorithm and details on how to implement. Along with the Baum-Welch algorithm came along details on the Forward and Backward algorithms commonly used for finding the optimized values of Hidden Markov Models. I also learned a bit about the Vertibi algorithm which is used in finding the most likely path that led to a certain state in Hidden Markov Models. My python knowledge also increased a great deal, since I had to create a whole project found ground up and use python to interact with the operating system which I had never done before.

CONCLUSIONS

My project resulting in generating a profit on the stock market, which was its initial goal. The profit was over 26%. To improve upon this work a more advanced trading simulator could be created, or this prediction program could be used on the stock market while be supervised. Another improvement would be to have the set always re-training every few weeks with recent data, as opposed to data that is years old and may reflect trends no longer present. It would appear that Hidden Markov Models are good at modeling consistent trends that occur, which may sometimes be the case with high volume frequently traded stock, but it is not always the case. Further reading has suggested that other forms of AI such as neural networks[4] may have a higher accuracy rating in predicting stock prices in comparison to a Hidden Markov Model.

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